

Toward Seasonal Estuarine Salinity: Evidence from Boreal Peatlands

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Abstract

Work in environmental research has long relied on assumptions we test directly here. Longer observation windows narrow the confidence interval but do not shift the central estimate. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting.

1. Introduction

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement

was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions.

2. Related Work

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3. Materials and Methods

Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Neither of the alternative formulations improved on the simple additive model in our setting. Calibration drift over the trial period remained below the threshold set out in the protocol.

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4. Experimental Setup

Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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5. Results

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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